

Ethernet LAN switching (PART 1)



OSI Model – Physical Layer

7	Application
6	Presentation
5	Session
4	Transport
3	Network
2	Data Link
1	Physical

- Defines physical characteristics of the medium used to transfer data between devices.
- For example, voltage levels, maximum transmission distances, physical connectors, cable specifications, etc.
- Digital bits are converted into electrical (for wired connections) or radio (for wireless connections) signals.
- All of the information in Day 2's video (cables, pin layouts, etc.) is related to the Physical Layer.





OSI Model – Data Link Layer

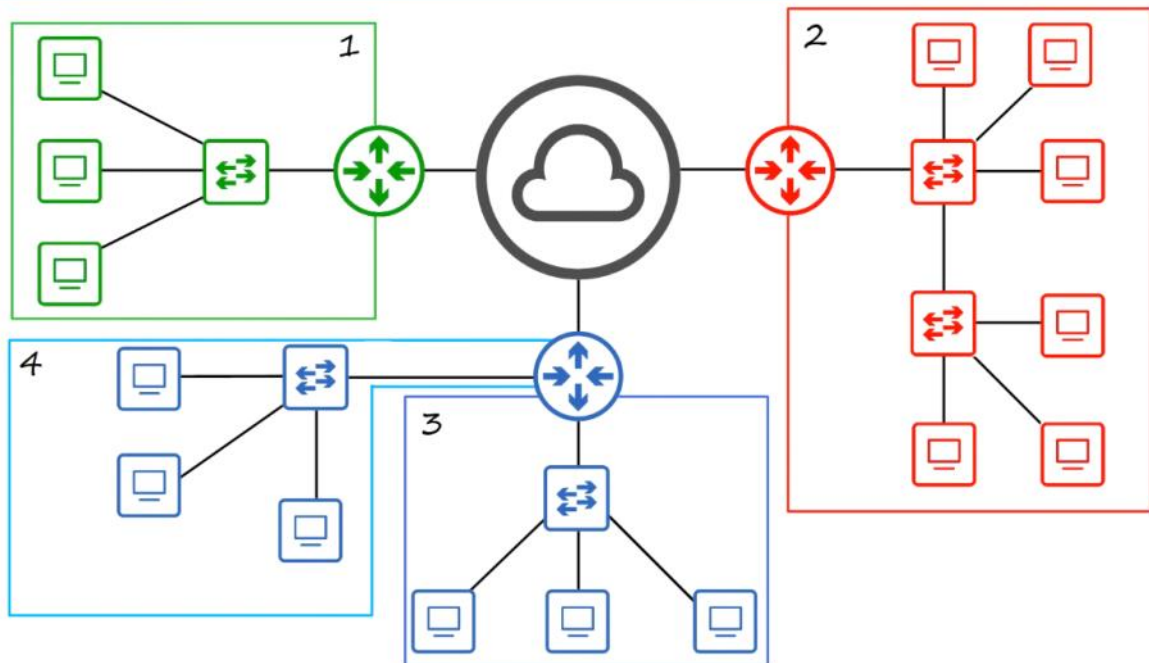
7	Application
6	Presentation
5	Session
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1	Physical

- Provides node-to-node connectivity and data transfer (for example, PC to switch, switch to router, router to router).
- Defines how data is formatted for transmission over a physical medium (for example, copper UTP cables)
- Detects and (possibly) corrects Physical Layer errors.
- Uses Layer 2 addressing, separate from Layer 3 addressing.
- Switches operate at Layer 2.





Local Area Networks (LANs)



A router has **multiple interfaces** (ports).

Each interface connects to a **different network**.

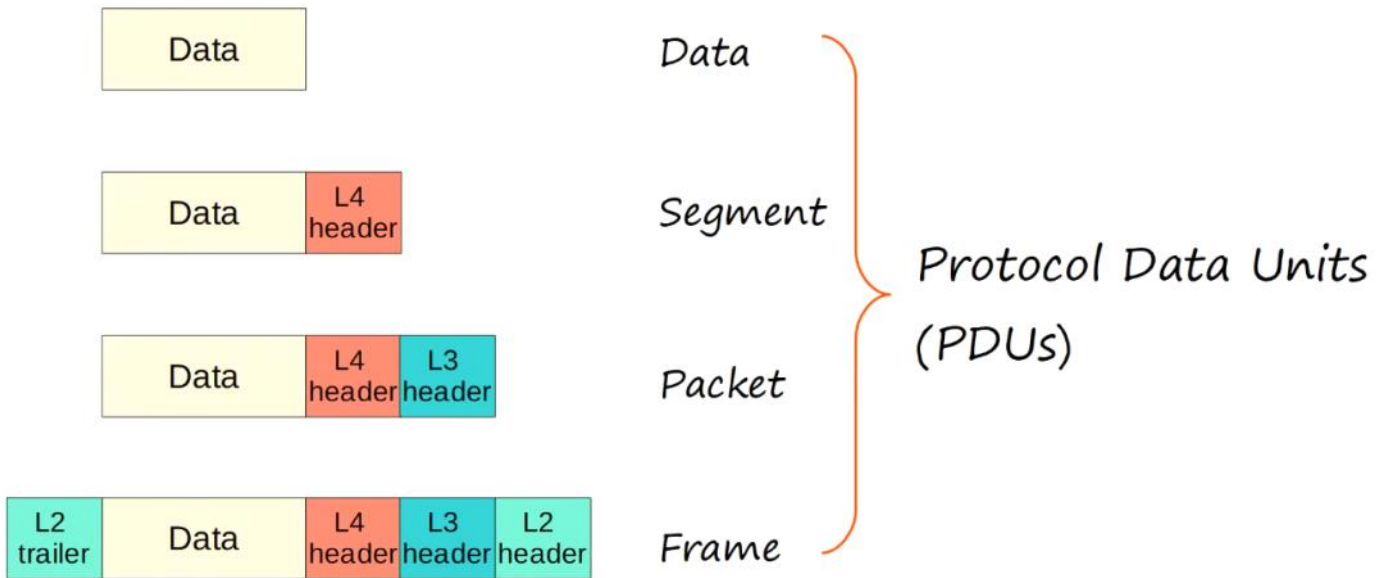
- Interface 1
- Interface 2
- Interface 3
- Interface 4

Each interface:

- Has its **own IP address**.
- Belongs to a **different subnet/network**.
- Acts as the **gateway** for that LAN.

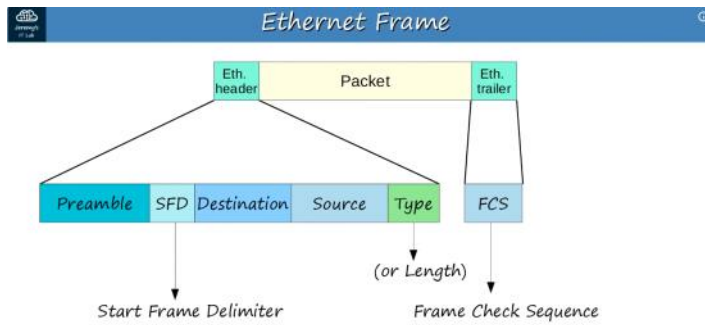


OSI Model – PDUs



Ethernet Frame

18 February 2026 06:04



Preamble

- Length: 7 bytes (56 bits)
- Alternating 1's and 0's
- 10101010 * 7
- Allows devices to synchronize their receiver clocks

SFD

- 'Start Frame Delimiter'
- Length: 1 byte (8 bits)
- 10101011
- Marks the end of the preamble, and the beginning of the rest of the frame

Destination & Source

Destination

- Indicate the devices sending and receiving the frame
- Consist of the destination and source 'MAC address'
- MAC = Media Access Control
- = 6 byte (48-bit) address of the physical device

Source

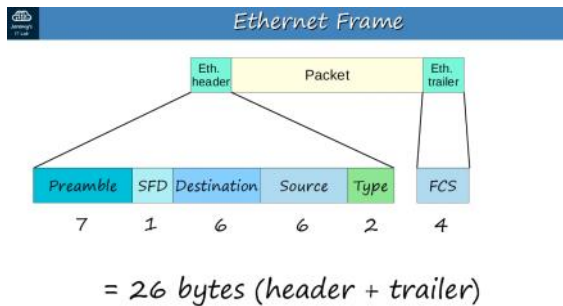
Type / Length

Type

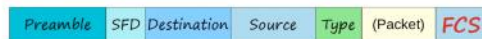
- 2 byte (16-bit) field
- A value of **1500 or less** in this field indicates the LENGTH of the encapsulated packet (in bytes)
- OR
- A value of **1536 or greater** in this field indicates the TYPE of the encapsulated packet (usually IPv4 or IPv6), and the length is determined via other methods

Length

IPv4 = 0x0800 (hexadecimal) IPv6 = 0x86DD (hexadecimal)
(2048 in decimal) (34525 in decimal)



Frame Check Sequence (FCS)



- 'Frame Check Sequence'
- 4 bytes (32 bits) in length
- Detects corrupted data by running a 'CRC' algorithm over the received data
- CRC = 'Cyclic Redundancy Check'



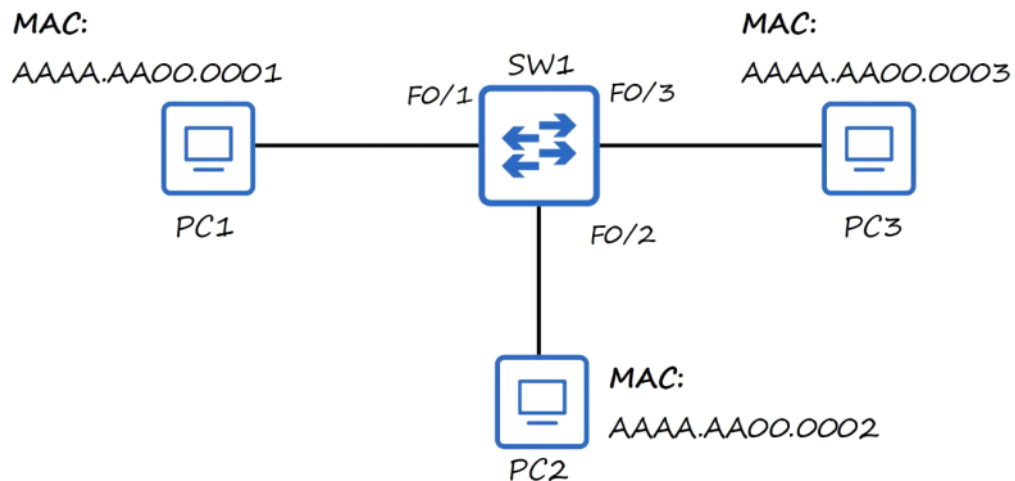
MAC Address

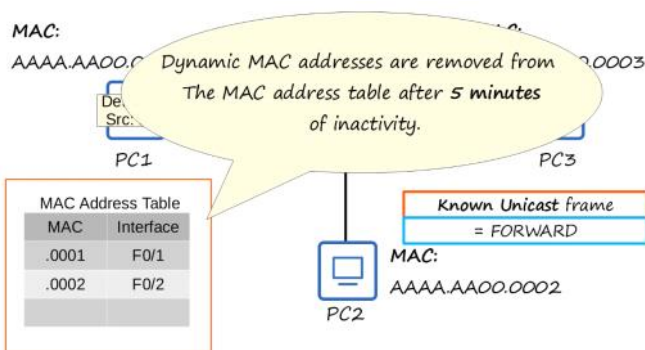
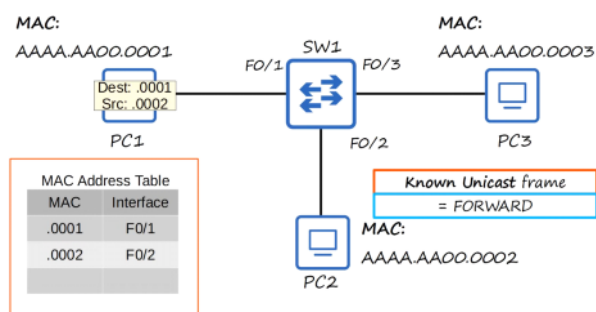
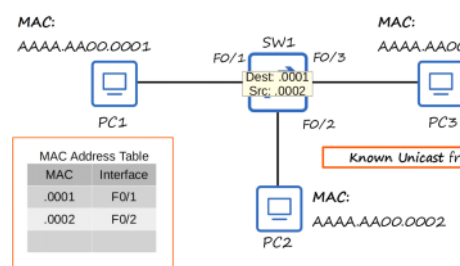
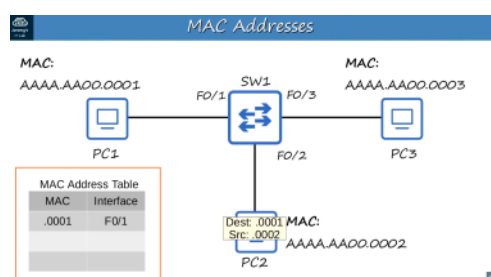
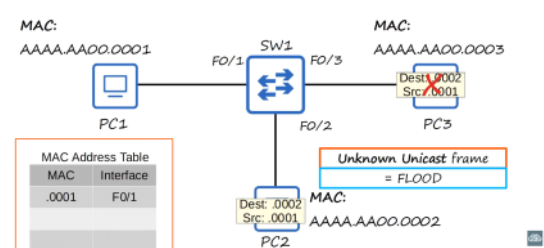
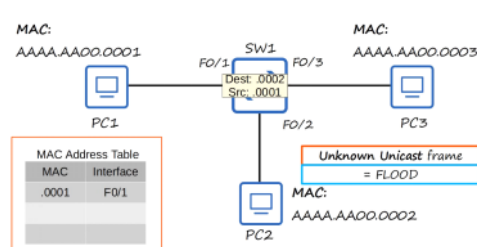
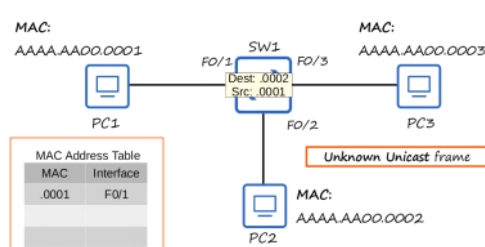
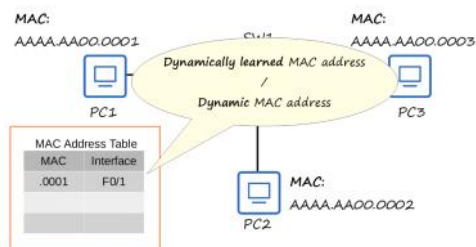
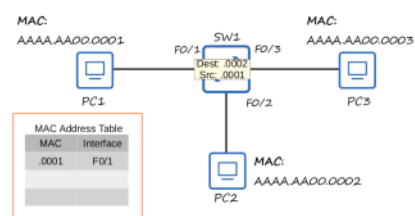
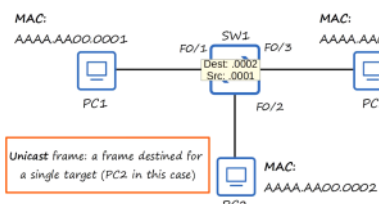
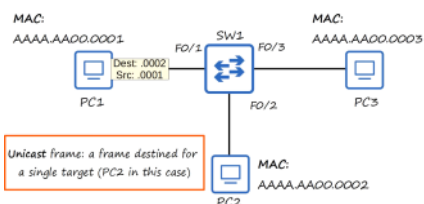
- 6-byte (48-bit) physical address assigned to the device when it is made
- A.K.A. 'Burned-In Address' (BIA)
- Is globally unique
- The first 3 bytes are the OUI (Organizationally Unique Identifier), which is assigned to the company making the device
- The last 3 bytes are unique to the device itself
- Written as 12 hexadecimal characters

Hexadecimal

DEC.	HEX.	DEC.	HEX.	DEC.	HEX.	DEC.	HEX.
0	0	8	8	16	10	24	18
1	1	9	9	17	11	25	19
2	2	10	A	18	12	26	1A
3	3	11	B	19	13	27	1B
4	4	12	C	20	14	28	1C
5	5	13	D	21	15	29	1D
6	6	14	E	22	16	30	1E
7	7	15	F	23	17	31	1F

00-0a-83-b1-c0-8e





An **unknown unicast frame** is an Ethernet frame where:

- The **destination MAC address is a specific device** (not broadcast/multicast)
- But the **switch does NOT know which port that MAC address belongs to**

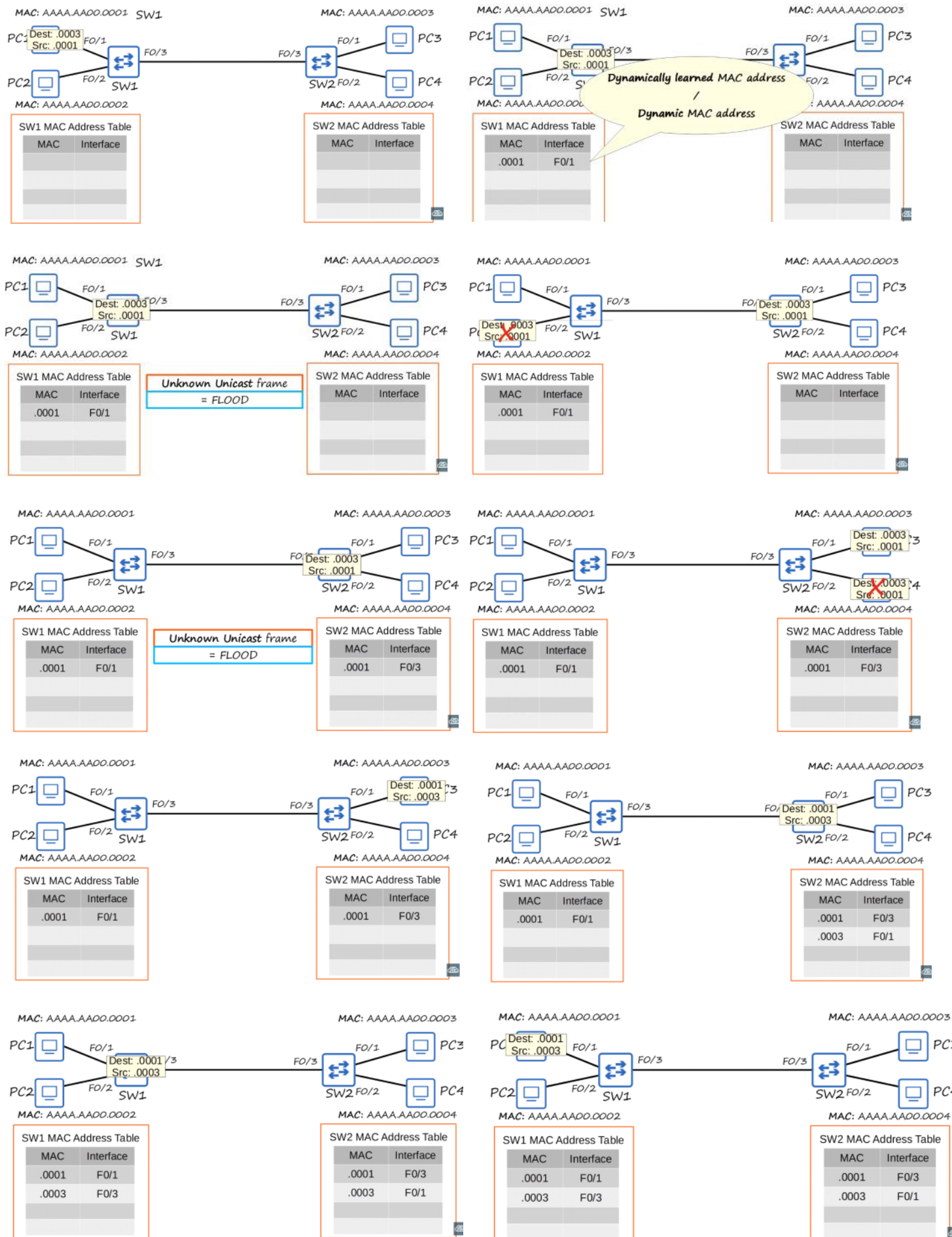
0.0003

2.0003

A **known unicast frame** is an Ethernet frame that is sent from one device to **one specific destination device**, and the switch **already knows** which port the destination device is connected to.

MAC learning & Frame forwarding

19 February 2026 00:49



MCQs

19 February 2026 01:32

Which field of an Ethernet frame provides receiver clock synchronization?

- a) Preamble
- b) SFD
- c) Type
- d) FCS

How long is the physical address of a network device?

- a) 32 bytes
- b) 32 bits
- c) 48 bytes
- d) 48 bits

What is the OUI of this MAC address? **E8BA.7011.2874**

- a) E8BA
- b) E8BA.70
- c) 7011
- d) E8BA.7011

Which field of an Ethernet frame does a switch use to populate its MAC address table?

- a) Preamble
- b) Length
- c) Source MAC Address
- d) Destination MAC Address

What kind of frame does a switch flood out of all interfaces except the one it was received on?

- a) Unknown unicast
- b) Known unicast
- c) Allcast

- A) Preamble
- D) 48 bits
- B) E8BA.70
- C) Source MAC Address
- A) Unknown unicast